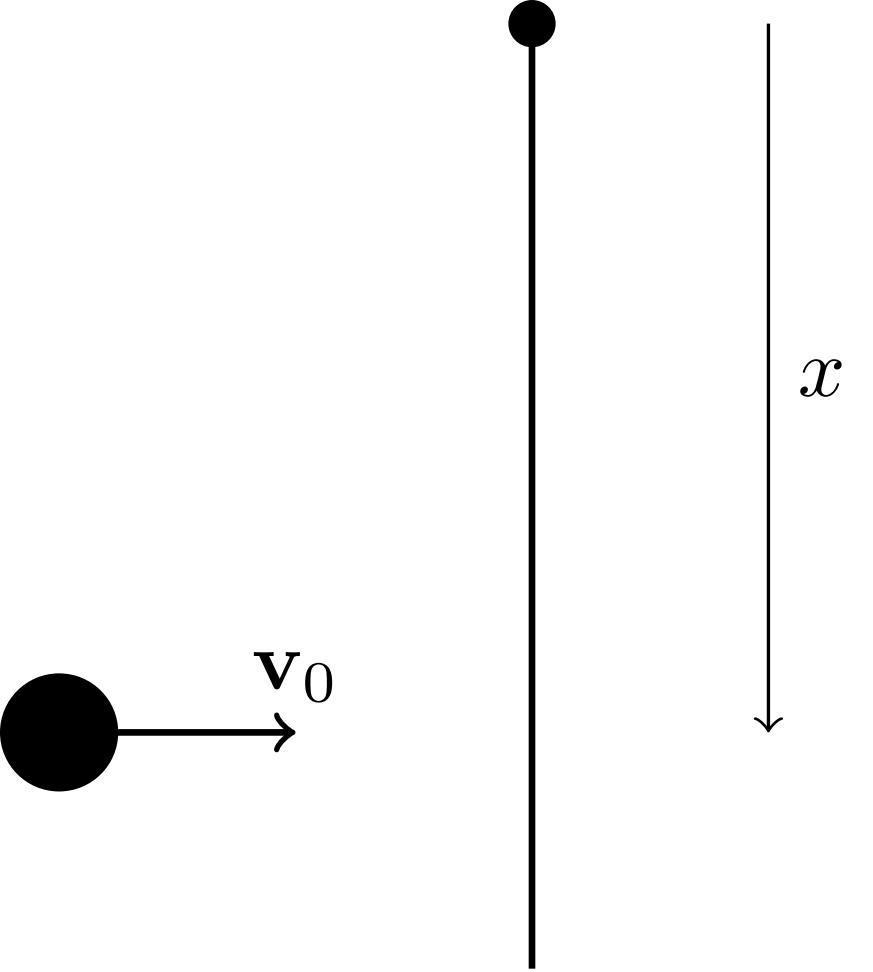
x
$\mathbf{v}_0$